

Comparative *In Silico* Analysis of Δ 8- and Δ 9-Tetrahydrocannabinol Binding Affinities at the Human Cannabinoid Receptor 1 (CB1)

Fabian A. Alvarez-Primo, Ph.D.

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ABSTRACT

This study provides a quantitative computational comparison of the binding thermodynamics between two primary isomers of tetrahydrocannabinol: Δ 8-THC and Δ 9-THC. Utilizing the crystal structure of the human Cannabinoid Receptor 1 (CB1), molecular docking simulations were conducted to determine the impact of double-bond positional isomerism on receptor affinity. Our results indicate that while both isomers exhibit spontaneous binding, Δ 8-THC demonstrates a slightly more favorable binding energy (-6.5 kcal/mol) compared to Δ 9-THC (-6.3 kcal/mol), suggesting that the shift of the double bond from the C9 to the C8 position facilitates a marginally optimized fit within the orthosteric hydrophobic pocket.

1. INTRODUCTION

The endocannabinoid system, primarily mediated by the CB1 receptor, plays a critical role in neuromodulation. Δ 9-Tetrahydrocannabinol (Δ 9-THC) is the most recognized phytocannabinoid; however, its isomer Δ 8-THC has gained significant attention due to its distinct legal and pharmacological profile. Despite their near-identical chemical formulas ($C_{21}H_{30}O_2$), the shift of a single double bond results in "significantly different" biological effects. This research seeks to quantify these differences through *in silico* molecular docking.

2. METHODOLOGY

Molecular structures were generated using **RDKit** and optimized with the MMFF94 force field.

OpenBabel was employed for PDBQT conversion. Docking was performed using **AutoDock Vina** targeting the **5TGZ** receptor structure. The search space was centered at (7.40, -1.60, 1.80).

3. RESULTS

Ligand Isomer	Binding Affinity (ΔG)	Configuration
$\Delta 8$ -Tetrahydrocannabinol	-6.5 kcal/mol	C8 Double Bond
$\Delta 9$ -Tetrahydrocannabinol	-6.3 kcal/mol	C9 Double Bond

4. DISCUSSION

The determination of a 0.2 kcal/mol affinity advantage for $\Delta 8$ -THC indicates that the positional change of the double bond alters the molecular geometry enough to improve van der Waals contacts within the CB1 receptor. This thermodynamic difference provides a structural foundation for the observed variations in psychotropic potency between the two compounds.

5. CONCLUSION

In conclusion, Dr. Alvarez-Primo's analysis confirms that $\Delta 8$ -THC is a marginally stronger binder to the CB1 receptor than $\Delta 9$ -THC in a static model. This underscores the sensitivity of G-protein coupled receptors to subtle ligand isomerism.